Diagonal Gaussian forcing: trace thresholds, completed divergence and recovery geometry

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Abstract

Countably many independent exponentially damped Gaussian modes define an actual square-integrable Hilbert-valued field at positive time exactly when the squared forcing amplitudes divided by the damping rates are summable. We prove this criterion, the exact covariance and cutoff error, and mean-square continuity up to time zero. The field is a closed linear Gaussian field in the original source space; its full Gaussian divergence exists, lies in second chaos, and has norm at most $\sqrt{2}$ times the field norm. The proof reconstructs finite Gaussian integration by parts and Hilbert–Schmidt completion, including the symmetric/antisymmetric kernel distinction. In contrast, observation of all Brownian modes at any positive time retains an infinite first-chaos isometry, preventing compact divergence recovery. Finite field energy and compact recovery are therefore separate questions. The trace and Gaussian mechanisms are classical; the paper consolidates frozen verified constructions and explicit counterexamples without claiming new stochastic PDE theory.

1 The actual modes and the trace criterion

Let $\mathcal{H}$ be a real separable infinite-dimensional Hilbert space with an isonormal Gaussian map $X : \mathcal{H} \to L^2(P)$: $\mathbb{E}X(h) = 0$, $\mathbb{E}[X(h)X(k)] = \langle h, k\rangle$, and finite families are jointly Gaussian. All scalar and field L^2 spaces below are over the Gaussian sigma-algebra generated by $X(\mathcal{H})$, modulo null sets; no additional non-Gaussian random factor is included. Embed mutually orthogonal copies of $L^2(\mathbb{R}_+)$ isometrically into $\mathcal{H}$, with embeddings ι_n, $n \geq 0$. Thus $B_n(t) = X(\iota_n \mathbf{1}_{(0,t]})$ are independent Brownian coordinates at the level of their joint Gaussian laws. Continuous versions give the usual Brownian realization if desired; only the stated L^2 coordinates are used below.

Fix deterministic rates $\lambda_n \geq 1$, amplitudes $c_n \in \mathbb{R}$, and an orthonormal sequence (e_n) in the same source coefficient space $\mathcal{H}$. Define

$$k_{n,t}(s) = \mathbf{1}_{(0,t]}(s) e^{-\lambda_n (t-s)}, \quad h_{n,t} = \iota_n k_{n,t}, \quad Y_n(t) = X(h_{n,t}). \tag{1}$$

The deterministic-integrand Brownian notation $Y_n(t) = \int_0^t e^{-\lambda_n(t-s)}\, dB_n(s)$ is equivalent to this isonormal definition. No adaptedness of the coefficient vectors e_n relative to the noise is needed. They are deterministic output directions; they need not be orthogonal to all input directions.

For $N \geq 1$ put

$$F_N(t) = \sum_{n<N} c_n e_n Y_n(t) \in L^2(P; \mathcal{H}).$$

This is a Hilbert-valued random field, not a scalar sum and not yet a claimed solution of an arbitrary SPDE.

Theorem 1.1. *For each $t > 0$, the following are equivalent:*

1. *$F_N(t)$ converges in $L^2(P; \mathcal{H})$;*
2. $\displaystyle\sum_{n\geq 0} \frac{c_n^2}{\lambda_n} < \infty.$

When these hold, write the limit as $F(t)$. Then

$$\|F(t)\|_2^2 = \sum_{n\geq 0} c_n^2 \frac{1 - e^{-2\lambda_n t}}{2\lambda_n}, \tag{2}$$

$$\|F(t) - F_N(t)\|_2^2 = \sum_{n\geq N} c_n^2 \frac{1 - e^{-2\lambda_n t}}{2\lambda_n} \leq \frac{1}{2} \sum_{n\geq N} \frac{c_n^2}{\lambda_n}. \tag{3}$$

Setting $F(0) = 0$, the curve $t \mapsto F(t)$ is continuous in $L^2(P; \mathcal{H})$ on $[0, \infty)$.

The reciprocal-rate condition is a diagonal instance of the classical stochastic-convolution Hilbert–Schmidt criterion; see Hairer's primary lecture notes [1], Sections 4.4 and 6. The theorem here fixes the actual source construction and distinguishes its positive-time quantifier from the trivial zero-time field.

2 Finite-mode identities and completion

Lemma 2.1. *For $\lambda > 0$, let $k_{\lambda,t}(r) = \mathbf{1}_{(0,t]}(r) e^{-\lambda(t-r)}$ and $V_\lambda(t) = (1 - e^{-2\lambda t})/(2\lambda)$. Then*

$$\|k_{\lambda,t}\|_{L^2}^2 = V_\lambda(t), \tag{4}$$

$$\langle k_{\lambda,t}, k_{\lambda,s} \rangle = e^{-\lambda|t-s|} V_\lambda(\min(s,t)), \tag{5}$$

$$\|k_{\lambda,t} - k_{\lambda,s}\|_{L^2}^2 = V_\lambda(t) + V_\lambda(s) - 2e^{-\lambda|t-s|} V_\lambda(\min(s,t)). \tag{6}$$

For $\lambda \geq 1$,

$$\frac{1 - e^{-2t}}{2\lambda} \leq V_\lambda(t) \leq \frac{1}{2\lambda}, \qquad \|k_{\lambda,t} - k_{\lambda,s}\|^2 \leq \frac{2}{\lambda}. \tag{7}$$

Proof. Integrating $e^{-2\lambda(t-r)}$ over $0 < r \leq t$ proves (4). If $s \leq t$, factor $e^{-\lambda(t-s)}$ from the overlap integral on $(0, s]$ and use (4); symmetry gives (5). Expansion of a squared norm proves (6). The first bounds follow from $0 \leq e^{-2\lambda t} \leq e^{-2t} \leq 1$. Finally

$$\|k_{\lambda,t} - k_{\lambda,s}\|^2 \leq 2\|k_{\lambda,t}\|^2 + 2\|k_{\lambda,s}\|^2 \leq 2/\lambda.$$

□

Proof of Theorem 1.1. Orthonormality of the deterministic output vectors gives, for any finite index set A,

$$\left\| \sum_{n\in A} c_n e_n Y_n(t) \right\|_2^2 = \sum_{n\in A} c_n^2 \mathbb{E} Y_n(t)^2 = \sum_{n\in A} c_n^2 V_{\lambda_n}(t). \tag{8}$$

In particular, disjoint finite index sets give orthogonal fields. Thus the partial sums are Cauchy exactly when the nonnegative energy series converges. At fixed $t > 0$, the factor $1 - e^{-2t}$ is positive, so (7) makes that convergence equivalent to the reciprocal-rate series. Completeness

gives $F(t)$. Passing finite orthogonal sums to their limits gives (2) and (3), including the stated upper error bound.

For continuity fix $s \geq 0$. Each right side in (6) is continuous in t and vanishes at $t = s$. Consequently $k_{\lambda_n,t} \to k_{\lambda_n,s}$ in L^2 for each n. Orthogonality gives

$$\|F(t) - F(s)\|_2^2 = \sum_{n\geq 0} c_n^2 \|k_{\lambda_n,t} - k_{\lambda_n,s}\|_{L^2}^2. \tag{9}$$

For example, this equality follows first for F_N, then by its convergence at t and s and the dominated summability in (7). The summable bound $2c_n^2/\lambda_n$ permits dominated convergence of the series, so the squared increment norm tends to zero. This also proves continuity at $s = 0$. □

When $c_n = 1$, the rates $(n+1)^2$ admit the field at all times, while $\lambda_n = n + 1$ do not admit the L^2 series at any positive time. Indeed the reciprocal square series converges, for example by comparison with the telescoping series $1/(n(n+1))$ for $n \geq 1$; the harmonic series diverges by grouping indices in dyadic blocks. For an undamped unit Brownian series the cutoff energy is Nt. Zero damping lies outside the hypothesis $\lambda_n \geq 1$ and is this separate elementary example, not a substitution into (2) with a zero denominator.

These statements assert L^2 convergence and mean-square continuity. They do not assert a samplewise uniform-in-time limit, a path-space Hölder exponent, or a continuum nonlinear PDE solution.

3 The original Gaussian divergence on linear fields

Let D be the original closed Gaussian gradient, obtained by closing the polynomial cylindrical gradient in $L^2(P) \to L^2(P;\mathcal{H})$. Its maximal Hilbert adjoint is the Gaussian divergence $\delta = D^*$. The relevant conventions agree with [2]. Write $\mathcal{L}$ for the closed linear span in $L^2(P;\mathcal{H})$ of fields $aX(b)$ with $a, b \in \mathcal{H}$. This is the space of linear Gaussian fields, not the full random field space.

Lemma 3.1. *For $a, b \in \mathcal{H}$,*

$$aX(b) \in D(\delta), \qquad \delta(aX(b)) = X(a)X(b) - \langle a, b\rangle. \tag{10}$$

For a finite-rank deterministic operator $K = \sum_{j=1}^m a_j \otimes b_j$, where $(a \otimes b)h = a\langle b, h\rangle$, put $F_K = \sum_j a_j X(b_j)$. Then

$$\|F_K\|_2^2 = \|K\|_{\mathrm{HS}}^2, \tag{11}$$

$$\|\delta F_K\|_2^2 = \|K\|_{\mathrm{HS}}^2 + \langle K, K^*\rangle_{\mathrm{HS}} = 2\|\operatorname{sym} K\|_{\mathrm{HS}}^2 \leq 2\|F_K\|_2^2. \tag{12}$$

Here $\operatorname{sym} K = (K + K^*)/2$. *The divergence is in second Gaussian chaos.*

Proof. For a polynomial cylindrical test f, Gaussian integration by parts applied to the product $fX(b)$ gives

$$\mathbb{E}[X(a)fX(b)] = \mathbb{E}[\langle Df, a\rangle X(b)] + \langle a, b\rangle \mathbb{E}f.$$

This proves (10) on the core. Passing graph approximants of f through the pairing extends it to every $f \in D(D)$; the right side of (10) is in L^2 by Gaussian moments. It therefore meets the defining maximal-adjoint bound.

The isonormal covariance gives $\mathbb{E}\langle F_K, F_L\rangle = \langle K, L\rangle_{\mathrm{HS}}$, proving (11) and representation independence. For the divergence, the centered Gaussian fourth-moment identity gives

$$\begin{aligned}\mathbb{E}[(X(a)X(b) - \langle a, b\rangle)(X(c)X(d) - \langle c, d\rangle)] \\ = \langle a, c\rangle\langle b, d\rangle + \langle a, d\rangle\langle b, c\rangle.\end{aligned}$$

That identity follows by differentiating the finite joint Gaussian exponential moment formula four times at zero. Summing proves the first equality of (12). Orthogonality of the symmetric and skew-adjoint parts in Hilbert–Schmidt inner product proves the second equality and the upper bound. The centered quadratic products are degree-two Hermite polynomials or their polarizations, hence lie in second chaos. □

Theorem 3.2. *Every field $F \in \mathcal{L}$ belongs to the original domain $D(\delta)$. The restricted divergence is bounded, with*

$$\|\delta F\|_2 \le \sqrt{2}\|F\|_2. \tag{13}$$

It maps into second chaos. For $F = F_K$ with K Hilbert–Schmidt, its kernel consists exactly of skew-adjoint K. Its canonical gradient recovery is $F_{\operatorname{sym} K}$ and

$$R(\delta F_K) = F_{\operatorname{sym} K}, \qquad \|\delta F_K\|_2^2 = 2\|F_{\operatorname{sym} K}\|_2^2. \tag{14}$$

Proof. The finite-rank isometry (11) extends by Hilbert–Schmidt completion onto $\mathcal{L}$. For finite-rank approximants $K_j \to K$, Lemma 3.1 shows that δF_{K_j} is Cauchy, with the bound (13). Closedness of the original adjoint therefore puts $(F_K, \lim_j \delta F_{K_j})$ in its original graph. Second chaos is closed, so the limit remains in it. The exact norm formula (12) passes to the limit, proving the kernel and energy assertions.

On centered Gaussian chaos define the canonical recovery as $R = DN^{-1}$, where the number operator N has eigenvalue q on chaos $q \ge 1$, and set $R1 = 0$. On second chaos it is $D/2$. Finite differentiation gives

$$D\delta F_K = F_{K+K^*} = 2F_{\operatorname{sym} K}$$

for finite-rank K. Passing to the original closed gradient graph proves the identity for Hilbert–Schmidt K, since both sides converge strongly. This gives (14). □

For clarity, the recovery used here is bounded on full scalar L^2: if $f = \sum_{q\ge 0} f_q$ is its orthogonal chaos decomposition, the finite Gaussian identity $\|Df_q\|_2^2 = q\|f_q\|_2^2$ gives

$$\|Rf\|_2^2 = \sum_{q\ge 1} \frac{\|f_q\|_2^2}{q} \le \|f\|_2^2.$$

The finite identity follows from the one-coordinate Hermite derivative and orthogonality, then tensorization. Different differentiated chaos orders are orthogonal. Thus the series defines R by norm completion. In the frozen library it is constructed as the minimal-norm original divergence recovery; its chaos identity identifies it with this DN^{-1}. Only its proved action in (14) and on first chaos is needed in this paper.

The constant $\sqrt{2}$ is attained for a unit $a = b$ in (10). Antisymmetric finite tensors instead give zero divergence with a nonzero field. Thus no isometry on all of $\mathcal{L}$ is asserted, and scalar divergence does not recover the antisymmetric component of a field.

4 Divergence and limits of the admitted forcing

Under the trace criterion of Theorem 1.1, define

$$K_t = \sum_{n\geq 0} c_n e_n \otimes h_{n,t}.$$

Orthonormality of e_n makes this an actual Hilbert–Schmidt sum: $\|K_t\|_{\mathrm{HS}}^2 = \sum_n c_n^2 \|h_{n,t}\|^2$. It follows that $F(t) = F_{K_t} \in \mathcal{L}$.

Corollary 4.1. *For every admitted forcing profile, $F(t) \in D(\delta)$ at every $t \geq 0$, and*

$$\delta F_N(t) \longrightarrow \delta F(t) \quad \text{in } L^2(P), \qquad \|\delta F(t) - \delta F_N(t)\|_2 \leq \sqrt{2}\|F(t) - F_N(t)\|_2.$$

The curve $t \mapsto \delta F(t)$ is mean-square continuous and has the exact symmetric recovery (14) *at each time.*

Proof. Apply Theorem 3.2 to $F(t)$ and to the differences of finite or limiting fields in $\mathcal{L}$. The norm estimate proves the convergence. Applying the same estimate to $F(t) - F(s)$ and using Theorem 1.1 proves continuity. Recovery follows from the identity for K_t. □

This divergence is the original Gaussian source adjoint, not an Itô integral in the physical time of F. In particular the field's deterministic output coefficient vectors may overlap its source directions, and the centered quadratic correction in (10) must be retained. The trace criterion controls the specific field; it does not make every observation or recovery map compact.

5 Infinitely many observed modes and noncompact recovery

Let the complete modal past at time t be

$$\mathcal{A}_t = \sigma(B_n(s) : n \geq 0,\ 0 \leq s \leq t), \qquad Q_t = \mathbb{E}[\,\cdot \mid \mathcal{A}_t].$$

These are the actual Gaussian observation spaces of all modes, independent of whether a particular amplitude/rate profile is trace-admitted.

Theorem 5.1. *The bounded operator $RQ_t : L^2(P) \to L^2(P;\mathcal{H})$ is compact if and only if $t = 0$. At every positive t, even recovery restricted to the observed first chaos contains an infinite-dimensional isometry.*

Proof. For $t > 0$ set $u_n = t^{-1/2}\iota_n \mathbf{1}_{(0,t]}$. These form an orthonormal sequence in $\mathcal{H}$, and $X(u_n)$ form an orthonormal sequence in $L^2(P)$. Each is observed at time t, so $Q_t X(u_n) = X(u_n)$. On first chaos the number operator is one and $DX(u_n) = u_n$, a deterministic coefficient vector. Therefore

$$RQ_t X(u_n) = u_n \quad \text{as a constant field in } L^2(P;\mathcal{H}).$$

These outputs are pairwise at distance $\sqrt{2}$. A compact operator would take this bounded input sequence to a sequence with a norm convergent subsequence, a contradiction. Taking closed linear spans also proves the asserted isometry. At $t = 0$ all modal observations are zero; Q_0 projects onto constants, which R kills. Thus $RQ_0 = 0$ is compact. □

The obstruction does not depend on the chosen canonical right inverse. Indeed, if a bounded linear map L satisfies $\delta L(X(u)) = X(u)$ for every u in an infinite observed coefficient subspace, adjointness against $X(v)$ gives

$$\langle \mathbb{E}L(X(u)), v\rangle = \mathbb{E}\langle L(X(u)), DX(v)\rangle = \langle u, v\rangle.$$

Thus expectation, followed by projection to that subspace, is a bounded left inverse to $u \mapsto L(X(u))$. Compactness of L would make the identity on an infinite-dimensional Hilbert space compact, contradicting its orthonormal sequence. This is the elementary left-inverse mechanism behind the frozen general recovery obstruction.

At zero damping, unit cutoffs have divergent energy; under quadratic damping, the particular field has finite energy and a valid divergence. Nevertheless the underlying observation family still contains every $B_n(t)$ and retains Theorem 5.1. The two conclusions concern different operators and do not contradict one another.

6 Formal correspondence and prior theory

The source authority is frozen verified commit `bab688bb6140cd6316c80680095a914ea0d346db`. The countable isonormal modes and actual orthogonal series are in `SPDECountableModes`; exponential profiles and reciprocal-rate criteria are in `SPDEExponentialForcing`. The actual Hilbert-valued field and divergence application are in `SPDEHilbertExponentialForcing`; time continuity is `SPDEExponentialTimeContinuity`. The completed original divergence, second-chaos and projection-energy results are `GaussianLinearFieldDivergenceCompletion`. The full observed mode obstruction is `SPDEObservationRecoveryCompactness`.
The theorem map distinguishes exact declarations from elementary human reorganizations: Hilbert–Schmidt tensors present the formal closed linear-field space, the finite fourth-moment proof reconstructs its quadratic completion, and cutoff errors follow from the exact orthogonal energy identities. The maximal Gaussian adjoint and coefficient-space geometry remain those of the frozen construction. No coefficient alignment is silently added to the application.
Hairer's primary notes establish the classical Hilbert–Schmidt stochastic-integral and stochastic-convolution setting. The scalar reciprocal-rate threshold, finite-chaos divergence and infinite-mode compactness obstruction are presented here as a consolidated formalization companion. Gaussian derivative/adjoint conventions are also compared with the primary treatment [2]. No new stochastic PDE existence theorem, optimal sample-path regularity, or priority claim is made. A samplewise space-time realization, nonlinear equation or stronger convergence topology requires its own hypotheses and proof. Full baseline build and axiom evidence was hash-checked rather than repeated at the expense of active research resources.

References

[1] M. Hairer, *An introduction to stochastic PDEs*, arXiv:0907.4178, version 2 (2023), Sections 4.4 and 6.

[2] T. Sottinen and L. Viitasaari, *Stochastic analysis of Gaussian processes via Fredholm representation*, Int. J. Stoch. Anal. (2016), article 8694365. doi:10.1155/2016/8694365.